

Designed pair: reachable, hazard-free, genuinely emitting
 $\text{core}(t_1) = \text{core}(t_2)$, $\text{gen}(t_1) = \text{gen}(t_2)$, $p_2 = (c_2, e_2)$

state t_1
 $\text{pay}(E_1) = [p_1, p_2]$
 p_2 already emitted

state t_2
 $\text{pay}(E_2) = [p_1]$
 p_2 still missing at c_2

for every obs with $\text{obs}(t_1) = \text{obs}(t_2)$,
every $P = g \circ \text{obs}$ has $P(t_1) = P(t_2) = B$

yes

$p_2 \in B?$

no

state t'_1
 p_2 appears twice
 $\text{duplicate}(t'_1)$
 $\Rightarrow \text{unsafe}(t'_1)$

state t'_2
 p_2 remains absent
 $\Rightarrow \text{lost}(c_2, t'_2)$

Either way, both repaired stores are clean at c_2 :
 $\text{clean}(t'_1, c_2) \wedge \text{clean}(t'_2, c_2)$